MILITARY SPECIFICATION SHEET

PUMP, VACUUM, POSITIVE DISPLACEMENT, AIR ROTARY VANE, TYPE 1, CAST IRON

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the pumps described herein shall consist of this document and the latest issue of Specification MIL-P-28824.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for type 1. rotary vane pump with cast iron housing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-320	-	Chromium Plating (Electrodeposited)

MILITARY

MIL-P-15024/5	-	Plate, Identification
MIL-E-15090	-	Enamel, Equipment, Light-gray (Formula No. 111.)
DOD-P-15328	-	Primer (Wash), Pretreatment, Blue (Formula No. 117-B for Metals)

STANDARDS

MILITARY

MS 33537	-	Helical Coil Installation Thread; Design Standard for

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with specification MIL-P-28824 and as specified herein.

3.2 Materials. The materials shall be as specified in Figures 2 through 9.

3.3 Details of components. The components shall be as specified in Figures 1 through 9.

3.4 Design and construction. The design and construction shall be as specified in Figures 1 through 9.

3.5 Dimensions. The dimensions shall be in accordance with Figures 3 through 9. The dimensions shall not vary outside the tolerances given within normal mean-time-between-failure (MTBF), except that surfaces in contact with carbon vanes may show expected wear characteristics only.

3.5.1 Alignment. Rotor alignment shall be checked by removing the end cap retainers and shims and checking the end play. The end play reading shall be at least .003 inches. The rotor shall be centered between end plates by shiming bearings.

3.5.2 Tolerances. Rotor and body tolerances shall be such that the rotor shall not come in contact with body or end plates whe operated at a temperature of 65°C.

3.6 Color and Finish.

3.6.1 External. The pump shall be primed in accordance with DOD-P-15328 and painted wih two coats of gray enamel conforming to type II, class 2 of MIL-E-15090.

3.6.2 Internal. Internal surfaces in contact with moving vanes shal' be chrome plated in conformance with Federal Specification QQ-C-320, class I1, type I. Plating shall be 0.0127 to 0.254 mm (0.0005 to 0.010 inches) thick and finished in accordance with Figures 3, 5, and 6.

3.7 Marking. Marking shall be in accordance with MIL-P-15024/5 to indicate tampering, and, if used, it shall contain a warranty note when warranty or other guarantee is stipulated in the contract. The warranty note shall be covered with clear lacquer and shall state, "This pump is under warranty until (month, year)". It shall not obstruct other required markings as shown in Figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Sampling and inspection shall be in accordance with specification MIL-P-28824 and as specified herein.

4.2 Teardown inspection. The teardown inspection shall, as a minimum, conform to the requirements of Appendix A. A teardown inspection shall be made as required and at the end of every reliability test. All dimensions and surfaces shall be checked to ensure that they are within the tolerances given in Figures 3 through 9.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation-packaging. Preparation for delivery shall be in accordance with specification MIL-P-28824.

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6. NOTES

6.1 The notes specified in specification MIL-P-28824 are also applicable to this specification.

6.2 Test equipment. The test equipment recommended for use in the tests given in Appendix A is listed in Table I of Appendix A.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
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(Proj. No. 4310-N097)

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APPENDIX A

INSPECTION PROCEDURES, REQUIREMENTS, AND TEST EQUIPMENT

10. SCOPE

10.1 Scope. This appendix details the inspection procedures for end housing, pump body, and pump rotor. It also includes inspection requirements and equipment.

20. INSPECTION PROCEDURES FOR END HOUSING.

20.1 Position air leaf guage head in seal bore of end housing and record the indicator reading.

20.2 Position air leaf guage head in bearing bore of end housing and record the indicator reading.

20.3 Remove air leaf guage head from bearing bore.

20.4 Measure the depth of bearing and seal shoulder from chrome surface with a depth guage and record readings.

20.5 Measure and record snap ring groove inside diameter depth and width.

20.6 Measure and record seal "O" ring groove inside diameter, depth and width.

20.7 Measure and record the end housing width and height.

20.8 Measure and record the diameter of the bearing end of the end housing.

20.9 Measure and record the bolt circle diameter for the mounting holes and bearing retainer.

20.10 Measure and record dowel pin radius.

20.11 Place end housing on a surface plate with the bearing end down and position air test indicator (ATI) gage head stylus on end housing chrome surface. Move stylus about surface and record ATI reading for flatness of Surface A.

20.12 Position end housing on truax fixture with Surface A down.

20.13 Position the ATI stylus on top of surface of bearing bore. Rotate end plate 360° and record total indicator reading (TIR).

20.14 Position the ATI guage head stylus on the inside top edge of bearing bore on side nearest ATI.

20.15 Rotate the end housing 360°. Record the difference of maximum and minimum readings.

20.16 Adjust ATI stylus downward to the bottom of the bore, just above the shoulder.

20.17 Repeat Step 20.14.

20.18 Reverse end housing on truax fixture.

20.19 Repeat Steps 20.13 through 20.16 for seal bore.

20.20 Measure finish with surface analyzer.

20.21 Check hardcoat for continuity and thickness.

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20.22 Check Helicoil installation in accordance with MS 33537.

30. INSPECTION PROCEDURES FOR PUMP BODY.

30.1 Set up body bore plunge jet guage head to masters (high and low limits) per instruction manual. (This shall be performed once daily prior to test.)

30.2 Set ATI to standard body width with the guage block on base surface of plunge jet gaging head. Readjust ATI column calibration as required.

30.3 Move ATI back from guage block and remove guage block.

30.4 Place pump body on plunge jet guage head.

30.5 Place ATI stylus on face of pump body. Rotate body 360° and record maximum and minimum ATI readings for body width. The difference between these maximum and minimum readings is the parallelism of the two faces.

30.6 Remove ATI from contact with pump body.

30.7 Record Column 1 reading for body bore diameter.

30.8 Rotate body until the float in Column 2 reaches its highest position. Align upper black line on sliding indicator with the top of float.

30.9 Rotate body 360°. The float shall remain within the indicated tolerance for squareness.

30.10 Reverse body on fixture and repeat Steps 30.8 and 30.9.

30.11 Measure and record the body "O" ring, inside and outside diameters and depth on both the drive end and antidrive end.

30.12 Measure and record the foot length, height and width and the foot mounting hole length and width.

30.13 Measure and record the diameters of the foot mounting holes.

30.14 Measure and record the body height and overall hex.

30.15 Measure and record the body radius and bore height to center line.

30.16 Measure and record the diameter and threads of the inlet and outlet ports.

30.17 With the body positioned on a surface plate with Surface A up, position ATI stylus on Surface A.

30.18 Rotate the body 360° and record the flatness of Surface A.

30.19 Measure finish with surface analyzer.

30.20 Check hardcoat for continuity and thickness.

30.21 Check Helicoil installation in accordance with MS 33537.

40. INSPECTION PROCEDURE FOR PUMP ROTOR

40.1 Set up guaging equipment to masters (high and low limits) per instruction manual. (This shall be performe once daily prior to test).

40.2 Measure rotor diameter with the "C" snap air guage and record reading of Column 5.

40.3 Measure rotor width with the "C" snap air guage and record reading of Column 6.

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40.4 Measure rotor bearing journal diameters (drive and antidrive) with the "C" snap air guage and record readings of Column 7.

40.5 Measure rotor seal seat diameters (drive and antidrive) with the "C" snap air guage and record readings of Column 8.

40.6 Measure and record the total rotor shaft length and the short shaft length.

40.7 Measure and record the shaft diameter.

40.8 Measure and record the snap ring diameter and width.

40.9 Measure and record the seal journal length and bearing journal length.

40.10 Measure and record the overall length from bearing journal to bearing journal.

40.11 Measure and record the length and depth of the flat on the drive end of the rotor shaft.

40.12 Measure and record the distances on the shaft between the center drill indentations and the bearing journals.

40.13 Measure and record the slot width, depth and distance from center line.

40.14 Position rotor in V block on bearing journals.

40.15 Place ATI stylus on diameter of rotor at extreme drive end of rotor. Rotate the rotor 360° and record the TIR for rotor runout.

40.16 Repeat Step 40.15 for the antidrive end.

40.17 Place ATI stylus on seal journal of the drive end of the rotor. Rotate the rotor 360° and record the TIR for the seal journal runout.

40.18 Repeat Step 40.17 for the antidrive end.

40.19 Place ATI stylus on shaft on drive end of rotor. Rotate the rotor 360° and record the TIR for shaft runout.

40.20 Repeat Step 40.19 for the antidrive end.

40.21 Check markings, nameplate and warning decals on pump as per drawing.

50. INSPECTION REQUIREMENTS

50.1 Inspection equipment. The equipment utilized in this procedure and listed by make or model number carry no implication of preference, recommendation or approval by the Department of Defense for use. It is recognized that equivalent equipment produced by other manufacturers may be capable of equally satisfactory performance in this procedure. The inspection equipment and fixtures required for inspection of the vacuum pump are listed in Table I.

50.2 Inspection conditions. Inspection are to be made in a controlled environment at local atmospheric pressure and controlled temperature of 23°C±1. Relative humidity will be between 40 and 65 percent during measurements.

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TABLE I. Inspection equipment

Manufacturer and Model Number	Description	Minimum Use Specifications
Bendix Model M08SM	Precisionaire 8 column instrument	Column 1 5000:1 "O" center scale 2 20,000:1 No scale 3 & 4 5000:1 "O" center scale 5 & 6 2000:1 "O" center scale 7 & 8 5000:1 "O" bottom scale
Bendix Model M01-5M	Precisionaire 1 column instrument	5000:1 "O" center scale
Bendix No. 60053834	Airtest indicator kit	None
Bendix No. 60053827	Set up block master	None
Bendix No. 60108081	Gage stand	None
Bendix No. 60055402	2-3 inch Plunjet airsnap	#1 2.1185-2.1195 width #2 2.8065-2.8075 O.D.
Bendix No. 60087000	Fixed C-type air-snap	#1 0.5905-0.5907 O.D. #2 0.7491-0.7511 O.D.
Bendix No. 2472290-13	Blind type leaf jet spindle	1.3745-1.3755 I.D.
Bendix No. 24722290-14	Blind type leaf jet spindle	1.3745-1.3785 I.D.
Bendix No. 2472290-2	Plunjet gaging spindle	#1 3.2465-3.2565 I.D. #2 90° ±1 minute squareness of bore to face
Bendix No. 2472290-20	Double diameter truax arbor assembly	As specified
Bendix AGD, Style #3 Class X	Master setting disc	Min. size 0.5905 dia. Max. size 0.5907 dia.
Bendix AGD, Style #3 Class X	Master setting disc	Min. size 0.7491 dia. Max. size 0.7511 dia.
Bendix AGD, Style #3 Class X	Master setting disc	Min. size 2.1185 dia. Max. size 2.1195 dia.
Bendix AGD, Style #3 Class X	Master setting disc	Min. size 2.8065 dia. Max. size 2.8075 dia.
Bendix AGD, Class X	Master ring gage	Min. size 1.3745 dia. Max. size 1.3755 dia.

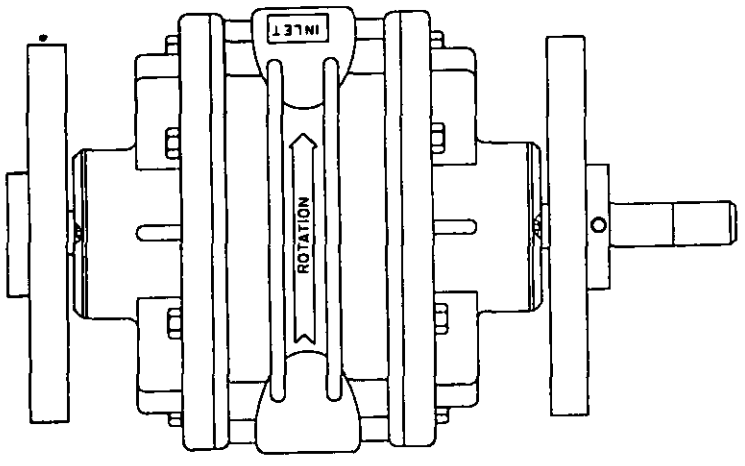
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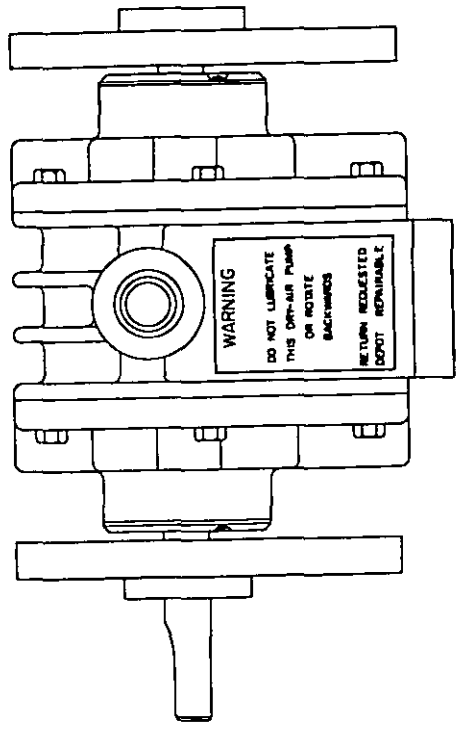
TABLE I. Inspection equipment

Manufacturer and Model Number	Description	Minimum Use Specifications
Bendix AGD, Class X	Master ring gage	Min. size 1.3775 dia. Max. size 1.3785 dia.
Bendix No. 24722290-3	Set of masters	Set-up masters for No. 2472290-2
Vernier Calipers		0-6 inch
Depth Micrometers		0-1 inch
Micrometers		0-1 inch
Micrometers		1-2 inch
Gage Blocks	Set	Do all Model 86-S

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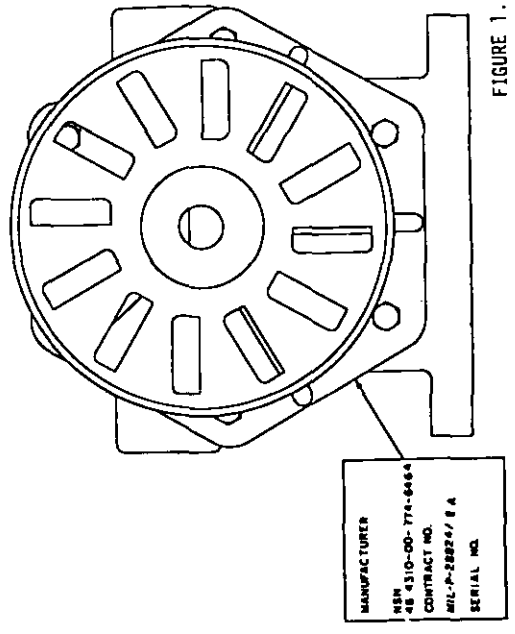
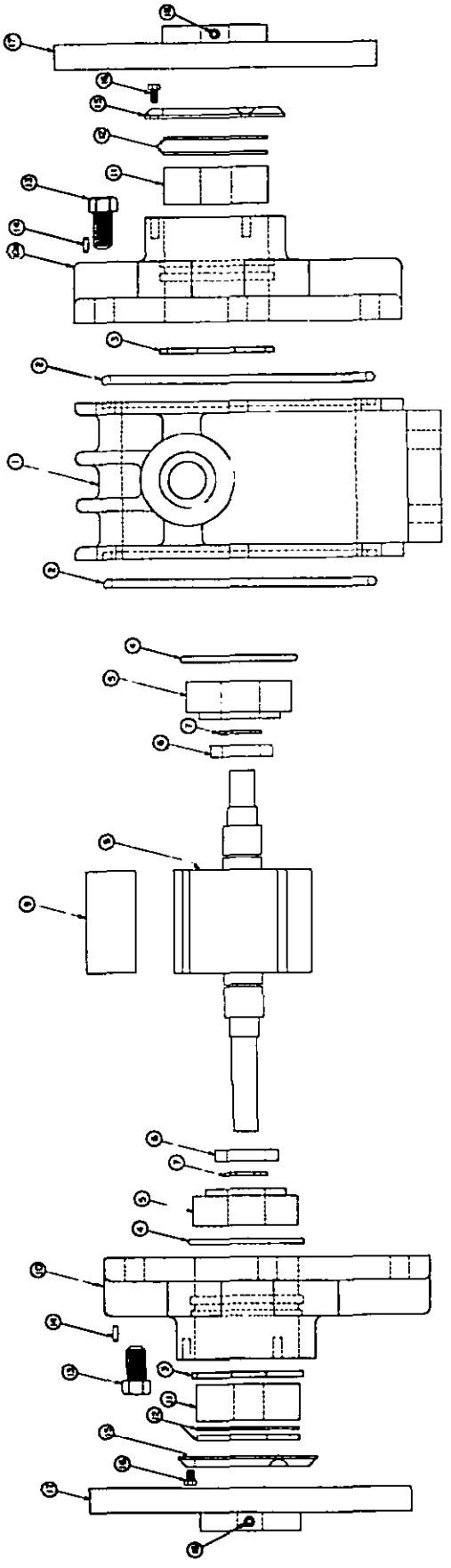


FIGURE 1. Pump assembly.

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DWG. P/N	NOMENCLATURE	QTY. REQ.	MATERIAL	DWG/SPEC OR NSN/MFG. NO.
1.	Main Pump Body	1	Cast Iron	
2.	"O" Ring	2	Rubber	5330-00-256-0201
3.	Snap Ring	2	Steel	Industrial Retaining Ring Co. #4000-137 or equiv.
4.	"O" Ring	2	Rubber	Achushnet Co. #568-126 or equiv.
5.	Seal (Main Body)	2	Steel/Carbon	6625-00-474-4083
6.	Seal (Carbon Ring)			
7.	Snap Ring	2	Steel	Industrial Retaining Ring
8.	Rotor	1	Stainless Steel Brass or Bronze	
9.	Vane	4	Carbon	

DWG. P/N	NOMENCLATURE	QTY. REQ.	MATERIAL	DWG/SPEC OR NSN/MFG. NO.
10.	End Housing	1	Cast Iron	
10a	End Housing	1	Cast Iron	
11.	Bearings	2	Steel	3110-00-944-0488 *
12.	Shims	As Req.	Steel	
13.	Bolt	12	Steel	1/4 - 20 x 5/8
14.	Dowel Pin	4	Steel	3/16 Dia. x 5/8 Lg.
15.	Bearing Retainer	2	Stainless Steel	
16.	Machine Screws 6-32 x 3/8	6	Steel	MS 35234-28
17.	Fan Assembly	2	Steel	
18.	Set Screw	2	Steel	1/4-20 x 3/8

* With the exception of lubricant, bearings identified by NSN 3110-00-944-0488 shall be used and shall be lubricated with MIL-G-81322. The amount of lubrication shall be 25 percent of void volume.

FIGURE 2. Pump exploded view.

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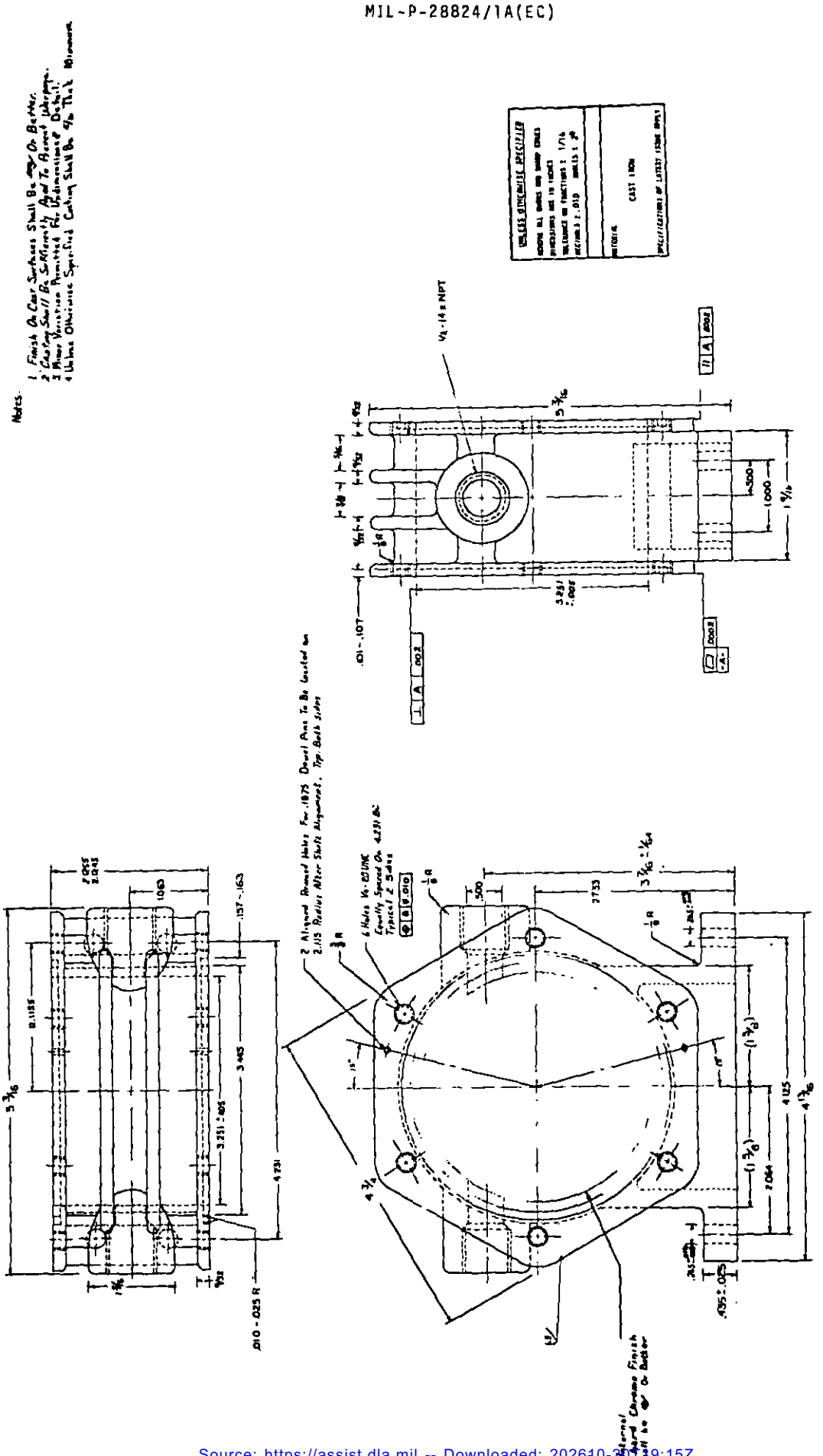


FIGURE 3. Main pump body.

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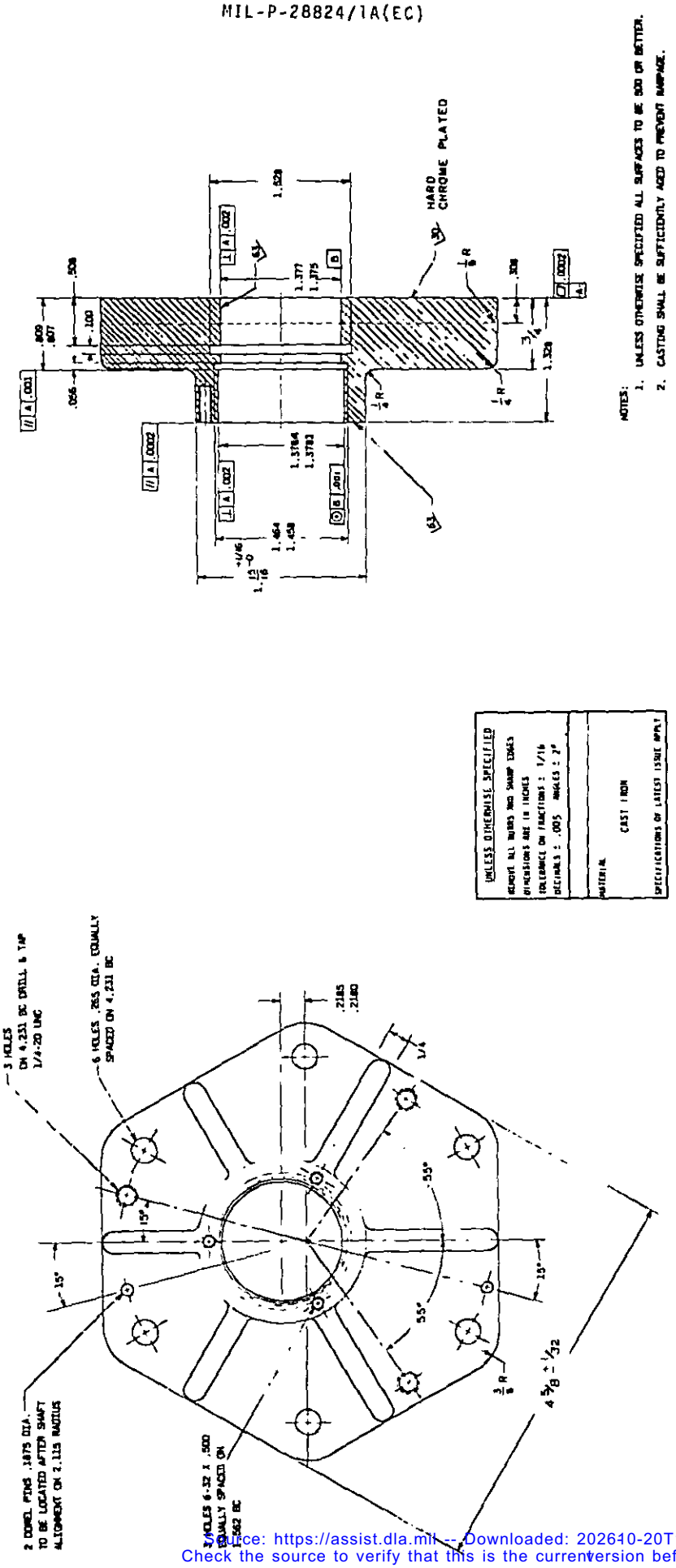


FIGURE 4. End housing.

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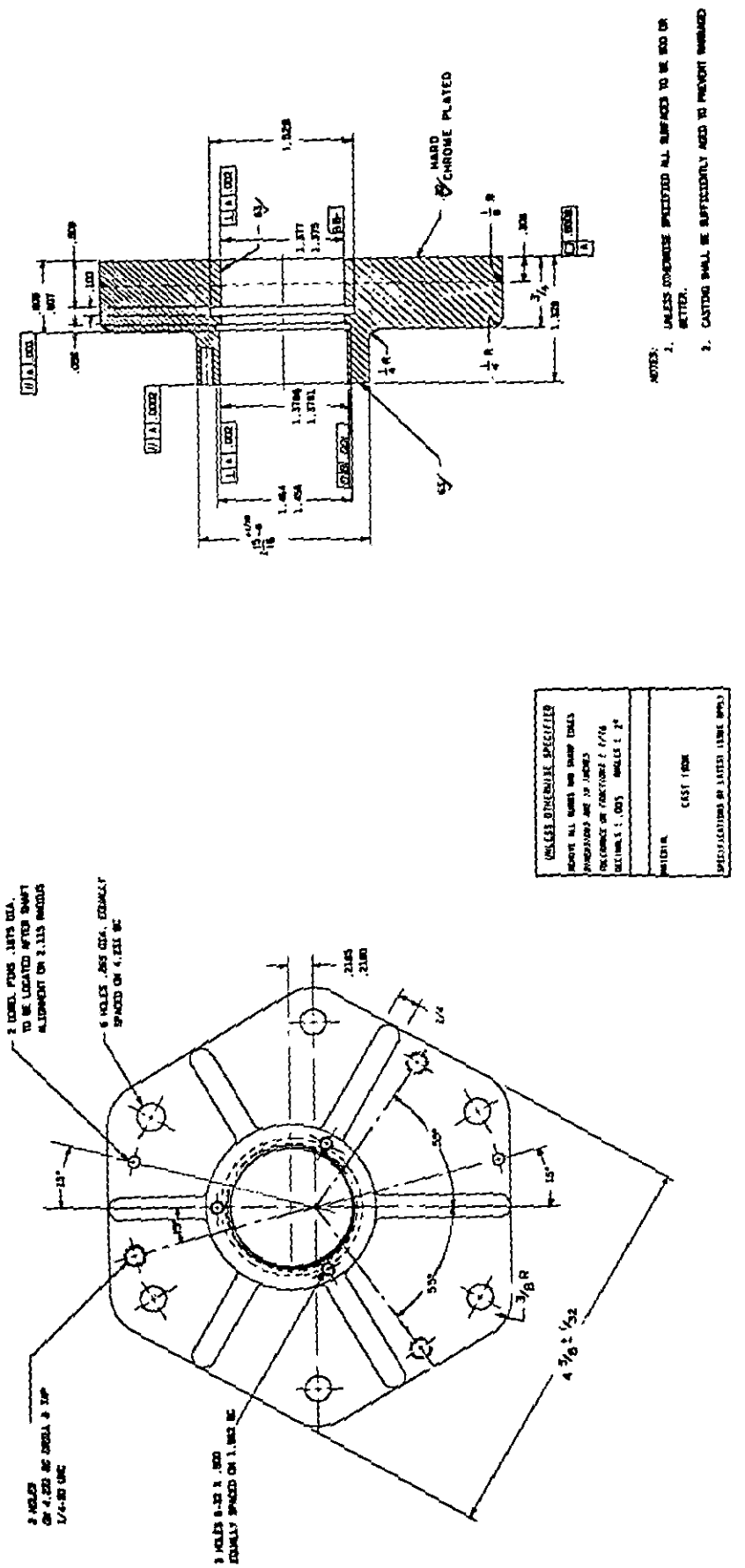


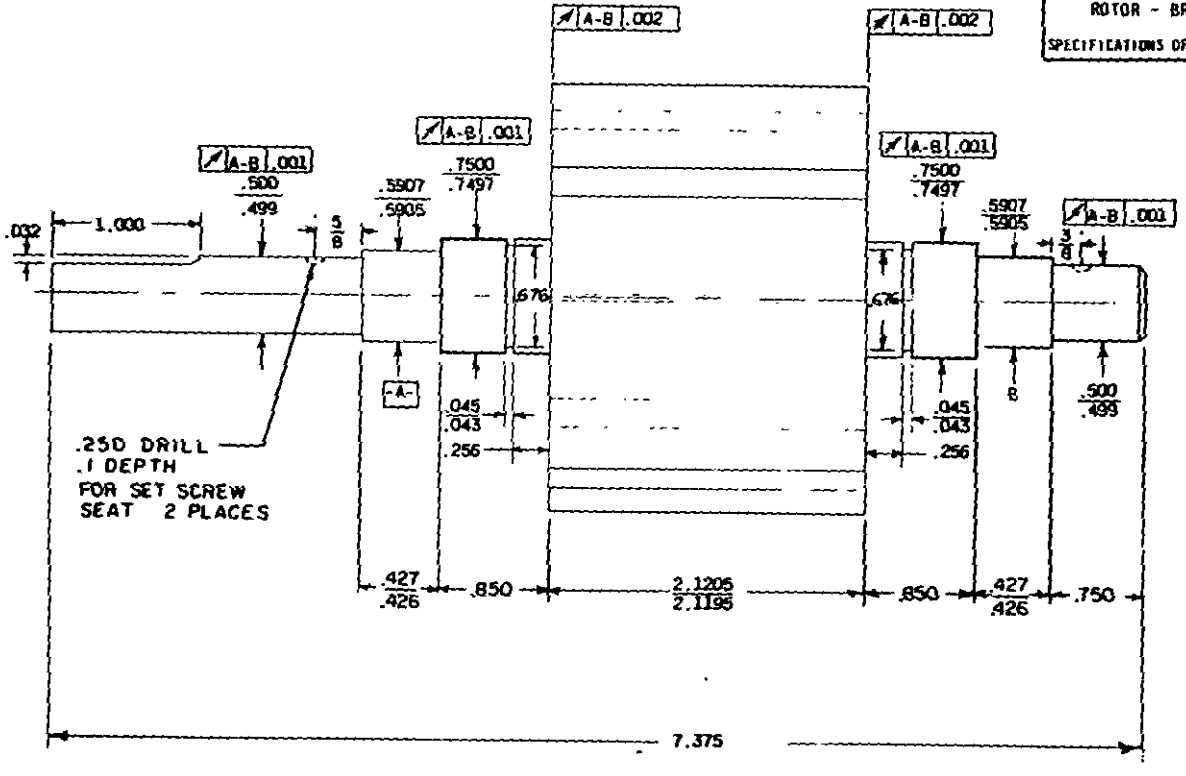
FIGURE 5. End housing.

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NOTES:

1. UNLESS OTHERWISE SPECIFIED ALL SURFACES SHALL BE 32/.
2. FACES OF [illegible] SHALL BE PARALLEL WITHIN .0005.
3. MATERIAL OF SHAFT TO BE ST. STEEL, ROTOR TO BE BRASS OR BRONZE OR ST. STEEL.
4. SHAFT AND ROTOR ARE TWO SEPARATE PIECES, ROTOR MAY BE CAST ON SHAFT OR ATTACHED TO SHAFT IN ANY ACCEPTABLE MANNER.

UNLESS OTHERWISE SPECIFIED
REMOVE ALL BURRS AND SHARP EDGES
DIMENSIONS ARE IN INCHES
TOLERANCE ON FRACTIONS ± 1/16
DECIMALS ± .005 ANGLES ± 2°
MATERIAL
SHAFT - STEEL
ROTOR - BRASS OR BRONZE
SPECIFICATIONS OF LATEST ISSUE APPLY



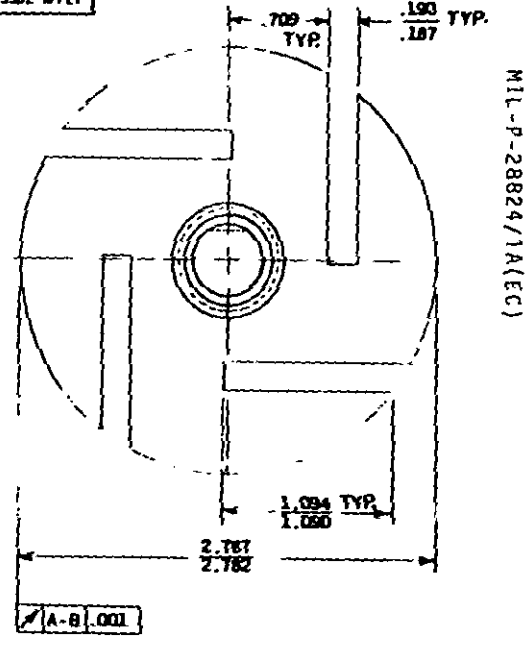


FIGURE 6. Rotor.

MIL-P-28824/1A(EC)

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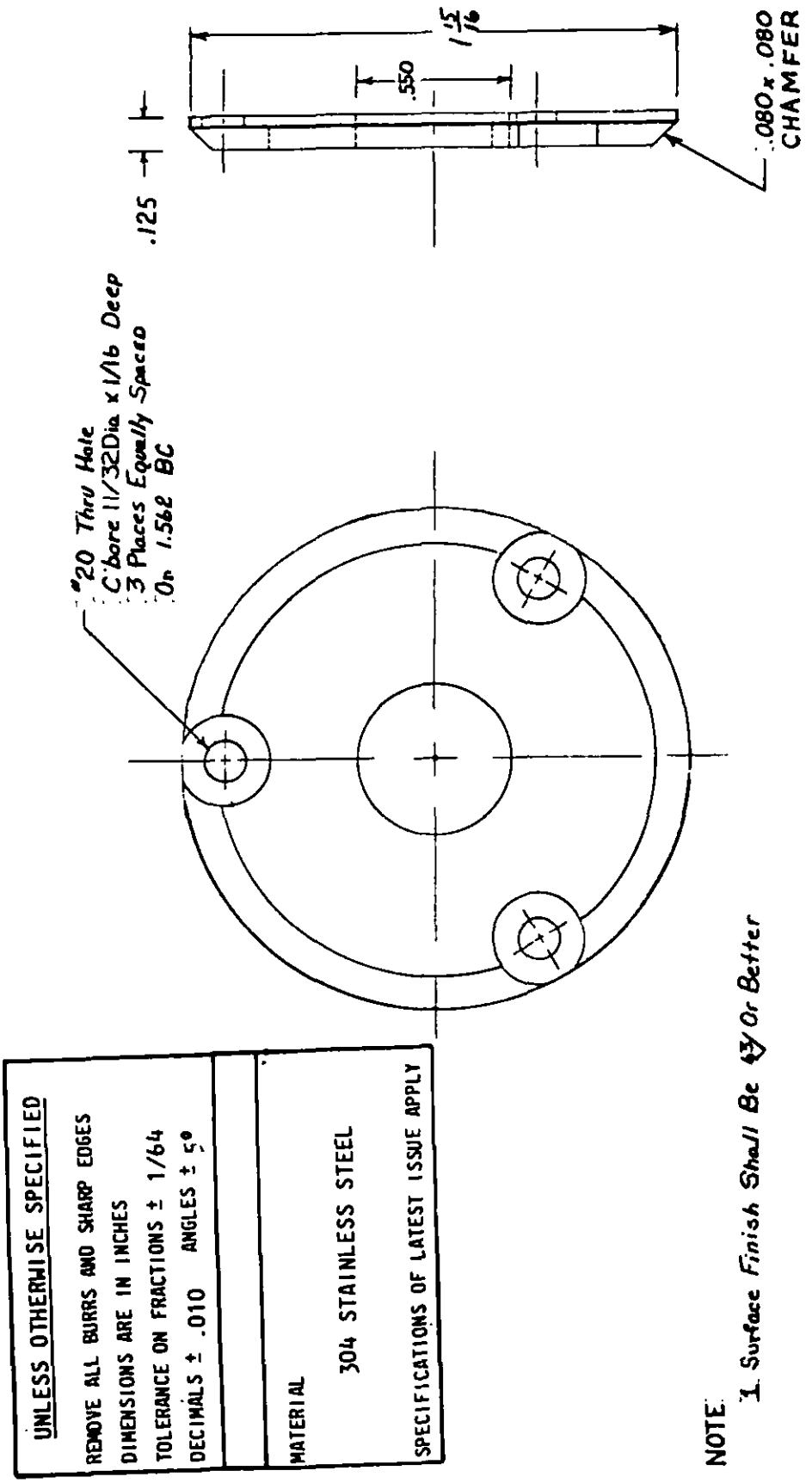


FIGURE 7. Bearing retainer.

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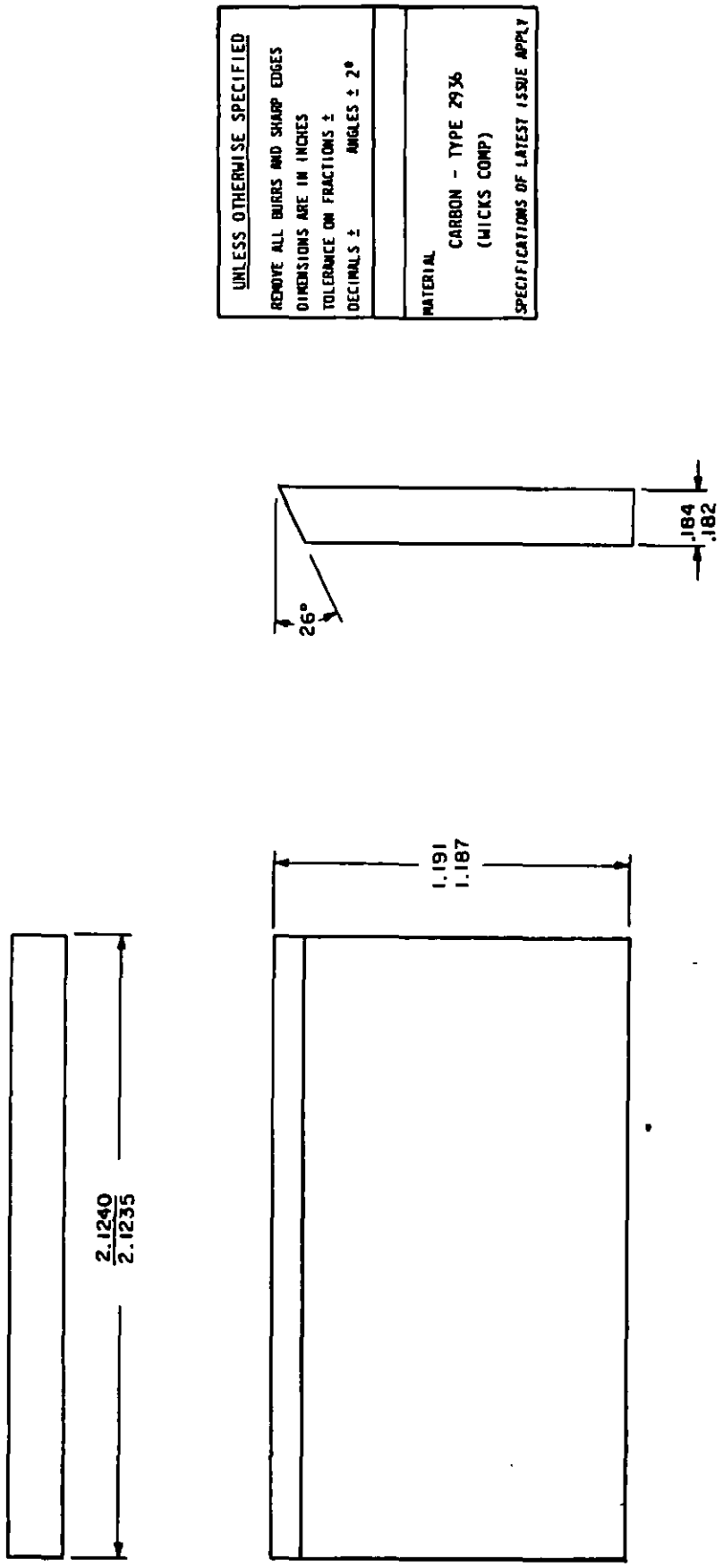
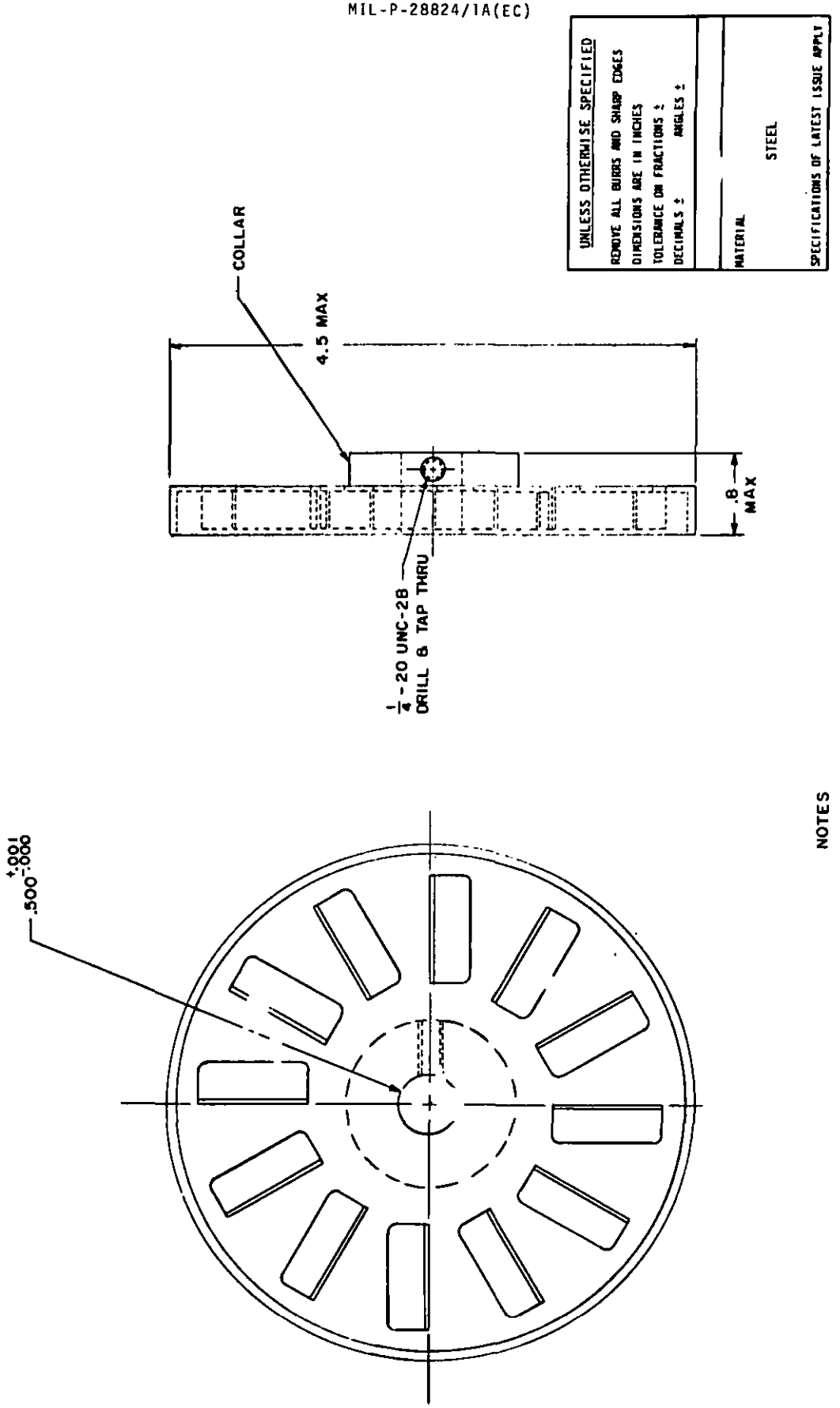


FIGURE 8. Vane.

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NOTES

1. FINISH SHALL BE $\sqrt{125}$ OR BETTER.
2. FAN DESIGN MAY BE OPTIONAL EXCEPT WHERE DIMENSIONS ARE GIVEN.

FIGURE 9. Fan.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-28824/1A(EC)	Pump, Vacuum, Positive Displacement, Air Rotary Vane, Type 1, Cast Iron

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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